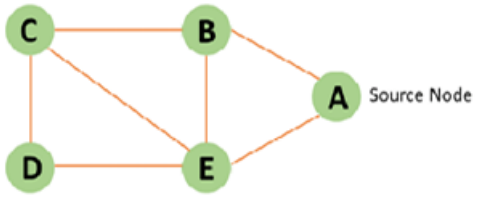
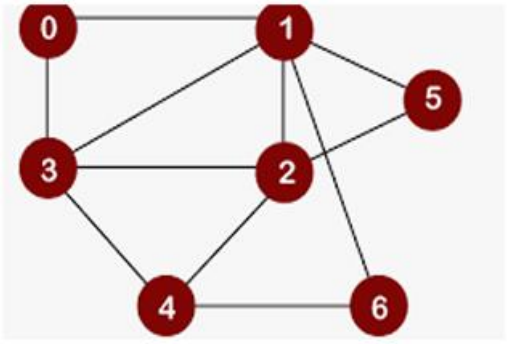
ADITYA UNIVERSITY

Department of freshman Engineering

Important Questions

Course name: Data Structures **Course code:** 2501CS03

UNIT-3					
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1		With neat diagrams, explain the following operations in Singly Linked List data structure. a) Insert element at the beginning b) Insert element at the end c) Delete the specified element d) Search the given element	L2	CO3	8M
2		Write the algorithm and explain the steps to perform the following operations on Doubly linked list. i) Add an element at the beginning of the list. ii) Delete an element at the beginning of the list. iii) Add an element after an item of the list. iv)Delete an element from the end of the list.	L2	CO3	8M
3	a	Explain how a single linked list can be reversed.	L2	CO3	4M
	b	List and explain the applications of linked list	L2	CO3	4M
4		Write the Pseudo code to perform all standard operations on Circular Linked List and explain the same with neat diagrams wherever necessary.	L2	CO3	8M
UNIT-4					
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1	a	Explain the implementation Stacks using array representation and devise the operations.	L2	CO4	4M
	b	Explain evaluation of post fix arithmetic expression using stack. Give the suitable example.	L3	CO4	4M
2	a	What is Queue? Write an algorithm for en-queue and de-queue operations and trace with an example.	L2	CO4	4M
	b	Write an algorithm for en-queue and de-queue operations of circular queue and trace with an example.	L2	CO4	4M
3	a	Explain in detail the list of operations that can be performed on a Circular Queues with appropriate diagrams.	L2	CO4	4M
	b	Explain any two types of queues with suitable examples.	L2	CO4	4M

4	a	How will you apply stack in converting the infix expression A+ (B*C (D/E^F)*G)*H into its equivalent postfix notation? Explain each step by tracing the stack contents.	L3	CO4	4M
	b	Write an algorithm for evaluating a Postfix expression using Stack data structure and illustrate the same by evaluating the following Postfix expression ABC*+D+, where A=2,B=3,C=4, and D=5.	L3	CO4	4M
		UNIT-5			
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1	a	Define binary tree? Explain the properties of binary tree	L2	CO5	4M
	b	Explain the insertion operation on binary search tree with suitable example	L3	CO5	4M
2	a	Construct the binary search tree for the following keys: 55, 42, 89, 34, 67, 78, 40, 22, 12, 56, 88, 65, 26, 98, 102	L2	CO5	4M
	b	Construct a Binary tree for the given Inorder and Postorder traversals by explaining the intermediate steps. Inorder Traversal is {4, 8, 2, 5, 1, 6, 3, 7} Post order Traversal is{8, 4, 5, 2, 6, 7, 3, 1}	L3	CO5	4M
3	a	Define Graph? Explain the two graph representation methods with suitable example.	L2	CO5	4M
	b	Which data structure is used in BFS implementation? Explain the step by step procedure to traverse the graph shown in figure 1 using BFS algorithm. 	L3	CO5	4M
4	a	Write an algorithm for DFS and BFS. Trace the algorithms for the following graph 	L2	CO5	8M